

Analysis of Android Ransomware

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1 Introduction

In this assignment, analysis of Android ransomware is performed using static and dynamic analysis. The Lab8 folder provided in the Windows XP image is utilized.

In the below figure SDK manager is opened and started latest device and The test Android is also started on the right side of the window.

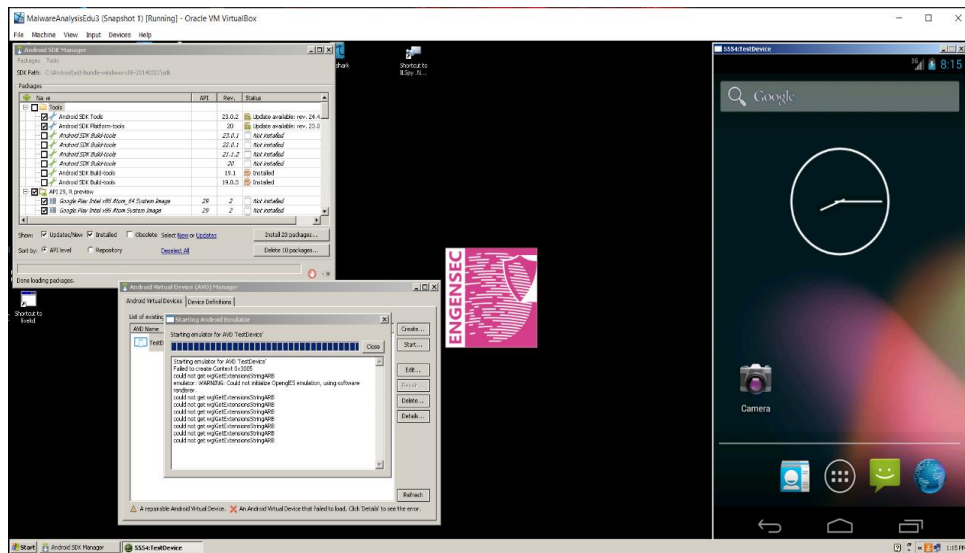


Figure 1: SDK Manager and Test Device

In the below figure, the Android Malware Analysis is identified and used for Further Analysis.

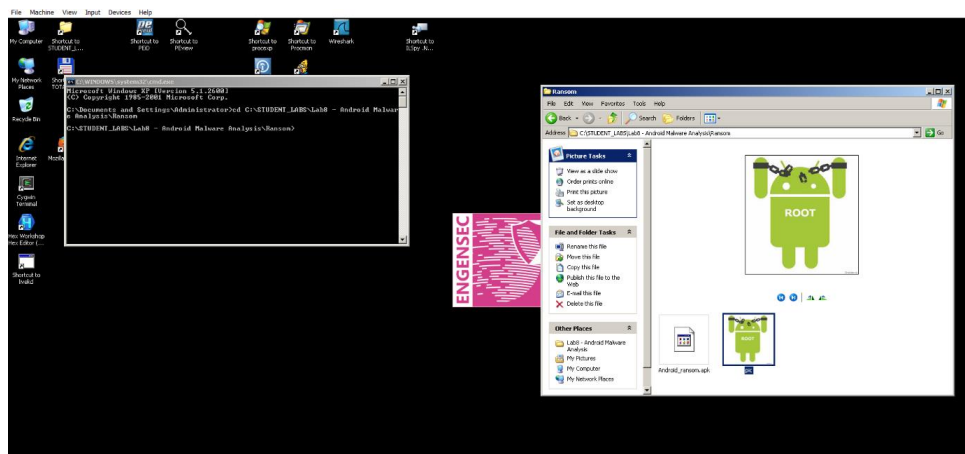


Figure 2: SDK Manager and Test Device

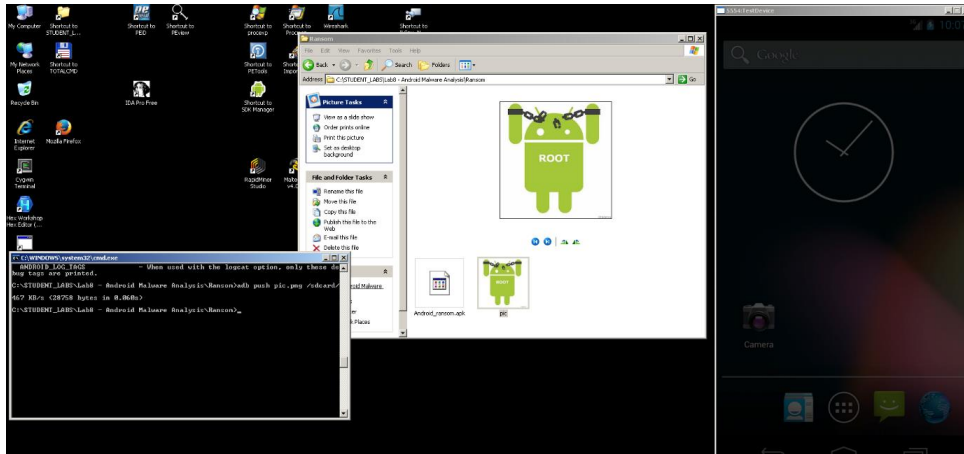


Figure 3: SDK Manager and Test Device

Now after setting location the image need to push to the SD card. Android_ransom.apk is also installed in that folder using the command "adb install Android_ransom.apk".

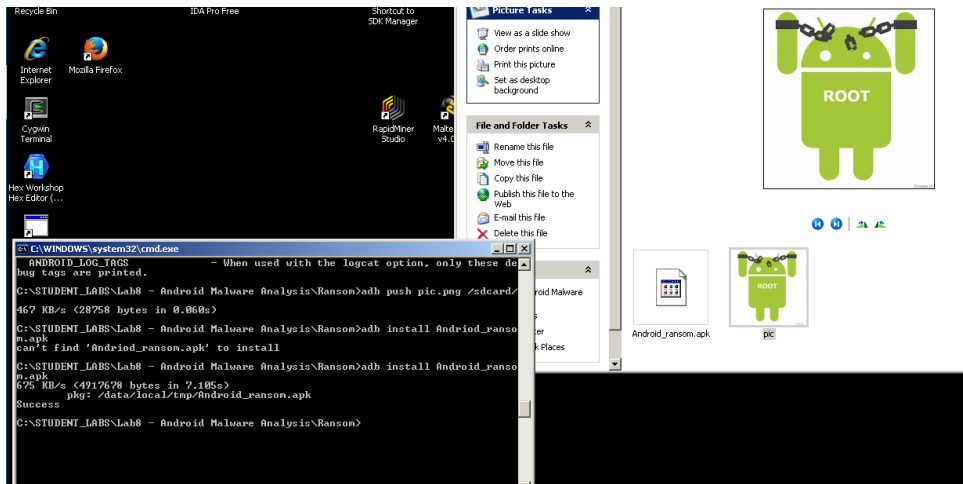


Figure 4: SDK Manager and Test Device

After installing we can see the device in the test device area. After installing the APK, when we click

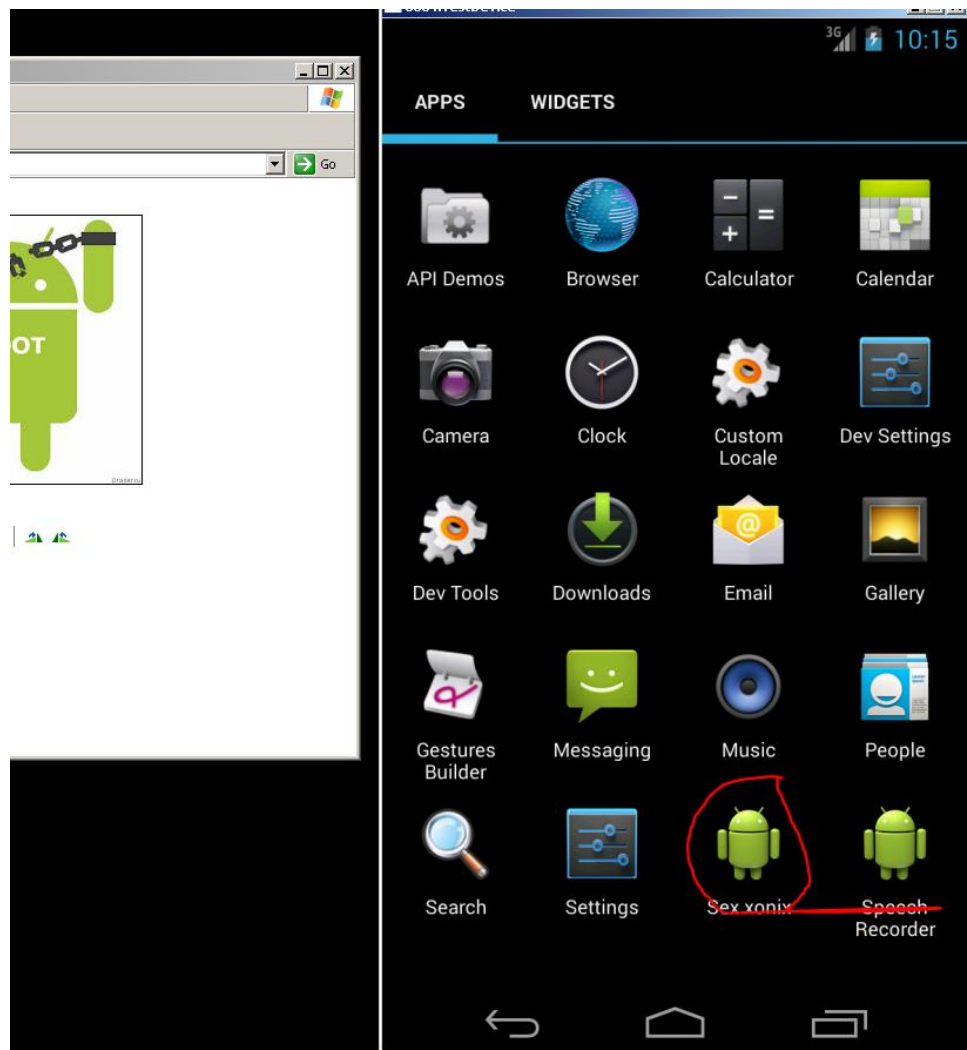


Figure 5: SDK Manager and Test Device

on the APK it displays a page which block us to not to go back until we click on the home button and application will not close. In the below figure we can see the displayed page after clicking apk on the right side on the picture.

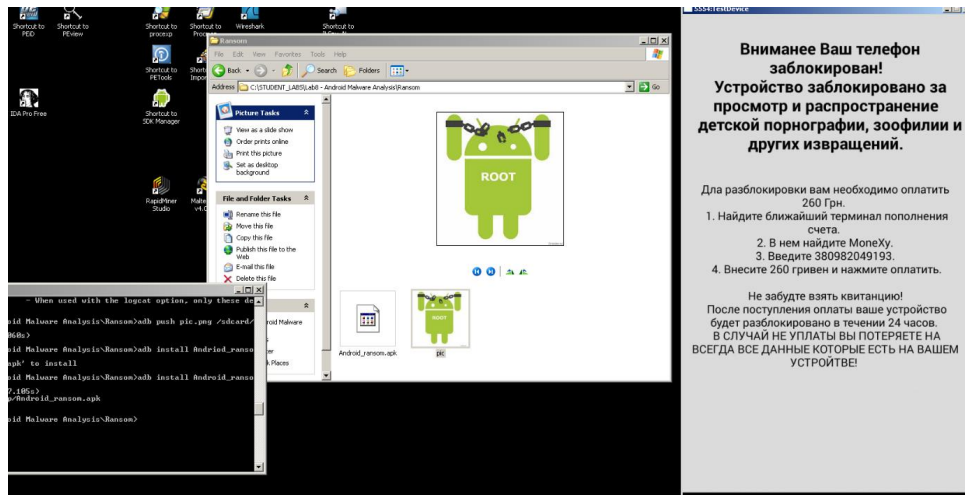


Figure 6: SDK Manager and Test Device

In the below picture packages are installed using command prompt by using the commands “adb shell” after this command you will enter into root@generic mode then enter the command “pm list packages -f”. In the picture all the packages can be seen on the left side.

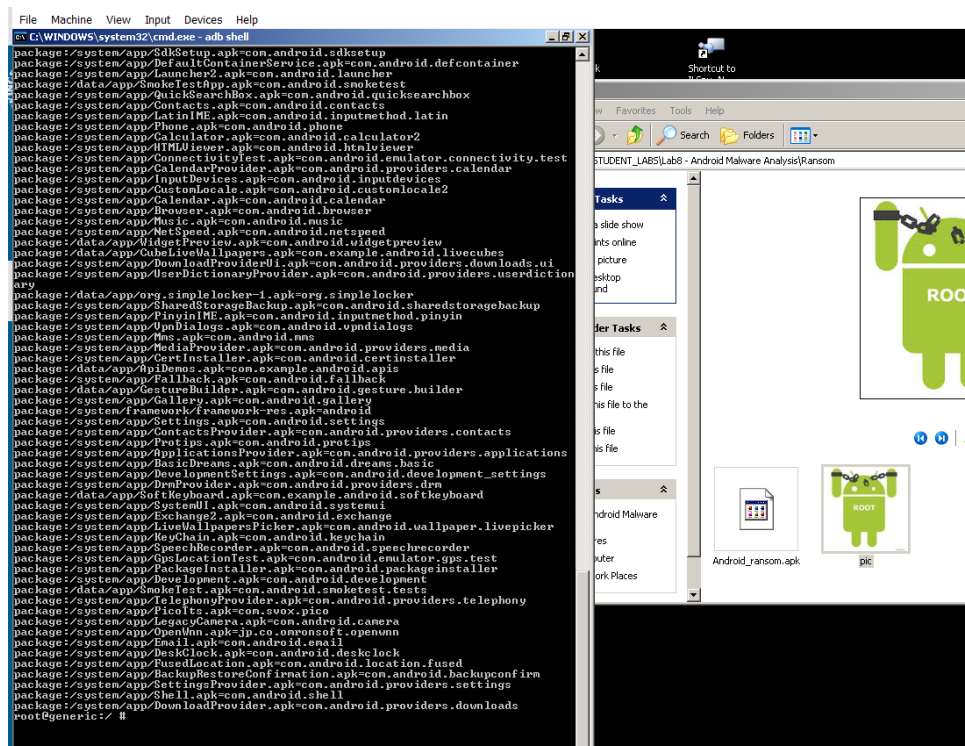


Figure 7: SDK Manager and Test Device

The previously copied file are check ed from the command prompt by entering the following command “adb shell”:- this command helps to run the command the emulator and after executing the command it enter into root@generic mode. “cd /sdcard/” :- changing the directory to sdcard. “ls” :- this command used to list the thing in that particular folder.

```
root@generic:/ # adb shell
adb shell
* daemon not running. starting it now on port 5038 *
* daemon started successfully *
root@generic:/ # cd /sdcard/
cd /sdcard/
root@generic:/sdcard # ls
ls
Alarms
Android
DCIM
Download
LOST.DIR
Movies
Music
Notifications
Pictures
Podcasts
Ringtones
pic.png.enc
root@generic:/sdcard #
```

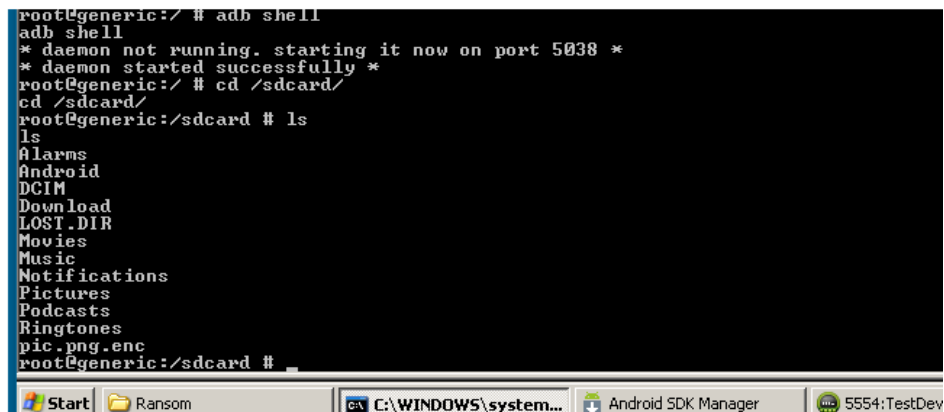


Figure 8: SDK Manager and Test Device

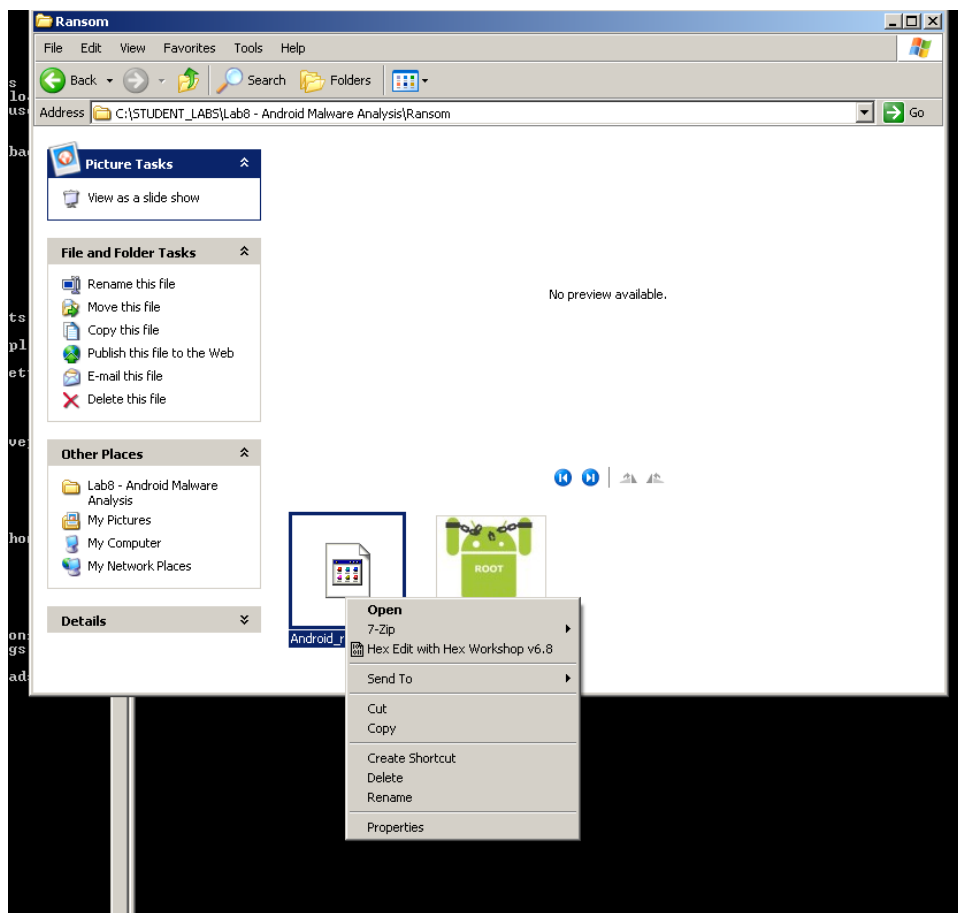


Figure 9: SDK Manager and Test Device

Now to convert the “.dex” to “.jar” we need to move to the folder called Android and then move to “dex2jar-0.0.9.9” then we can use the command “dex2jar.bat classes.dex”, this conversion will help in such a way that the java decompiler can read the “.jar” file.

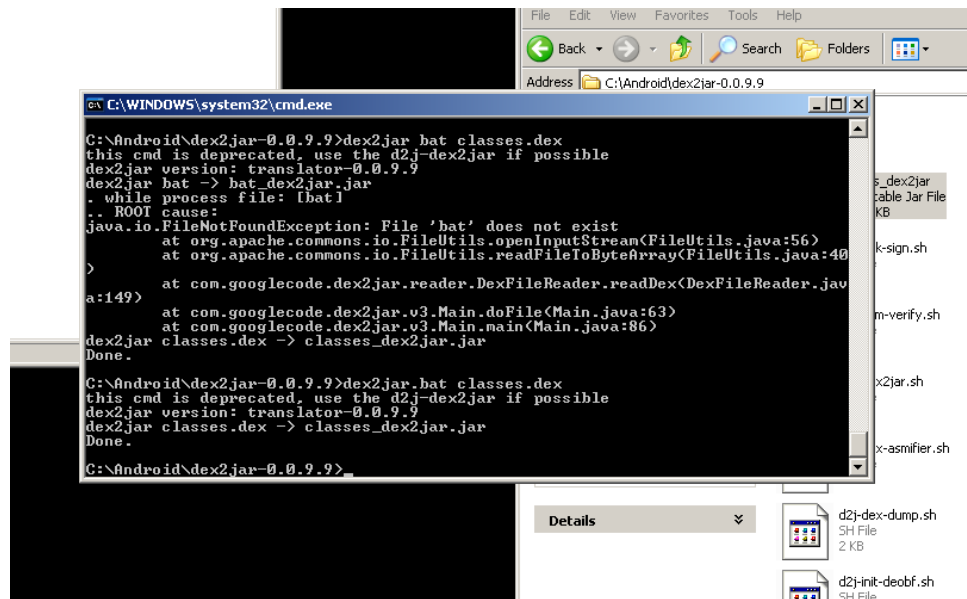


Figure 10: SDK Manager and Test Device

From here we will perform the Analysis in two different ways 1. Static analysis: without executing. 2. Dynamic analysis: with executing. Static Analysis By using java decompiler we will open the converted “.dex” file (jar file). When we open the Aesencrypt.class from simplelocker we can see the encryption algorithm as “SHA 256”. if we open the FilesEncryptor.class we can see the cypher key “jndlasf074h”. This encryption algorithm and cypher key can be seen in the below 2 pictures.

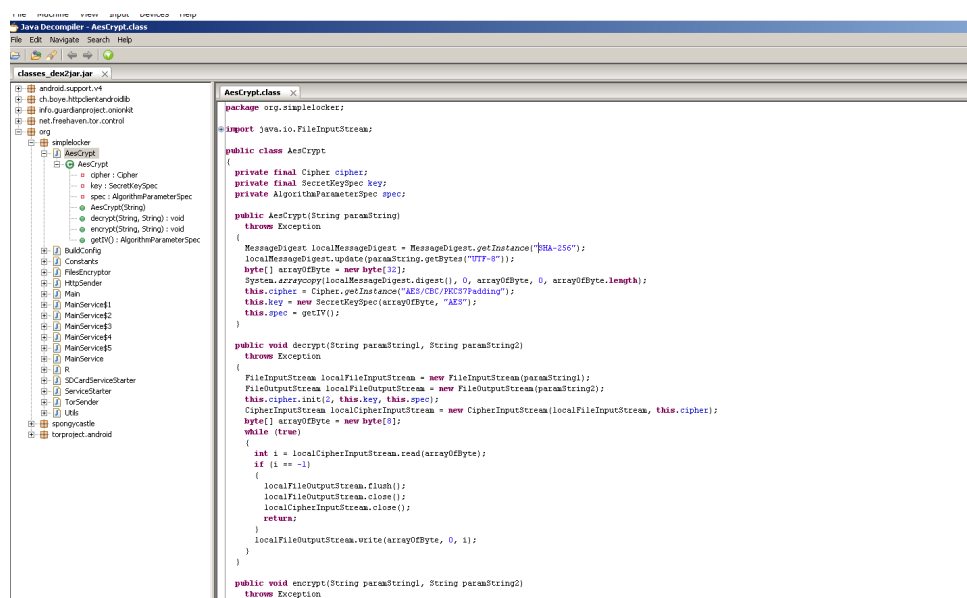


Figure 11: SDK Manager and Test Device

When we open “constants.class” file, we can see the ADMIN_UR which is decompiled and other URL are leading to a backdoor activity.

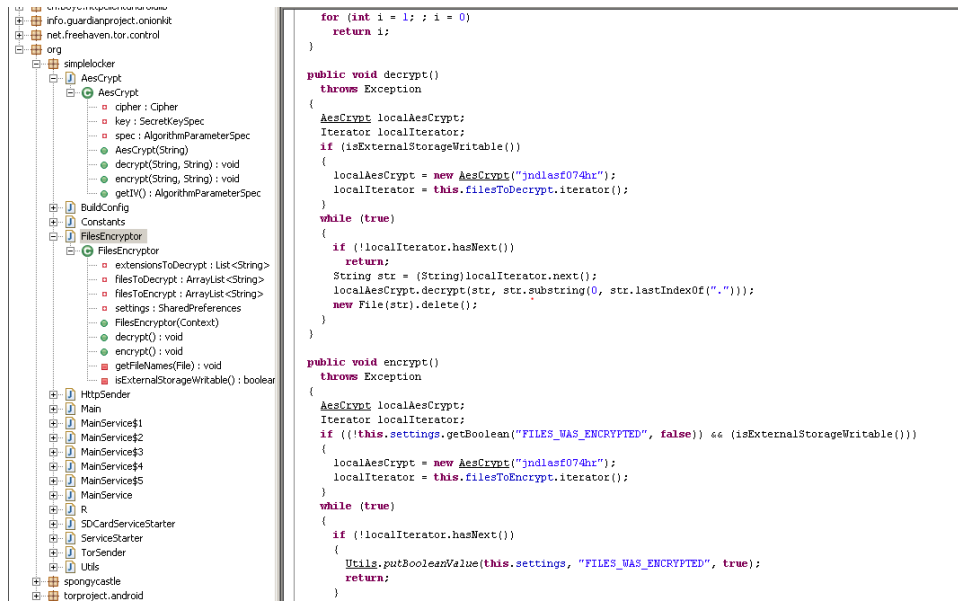


Figure 12: SDK Manager and Test Device

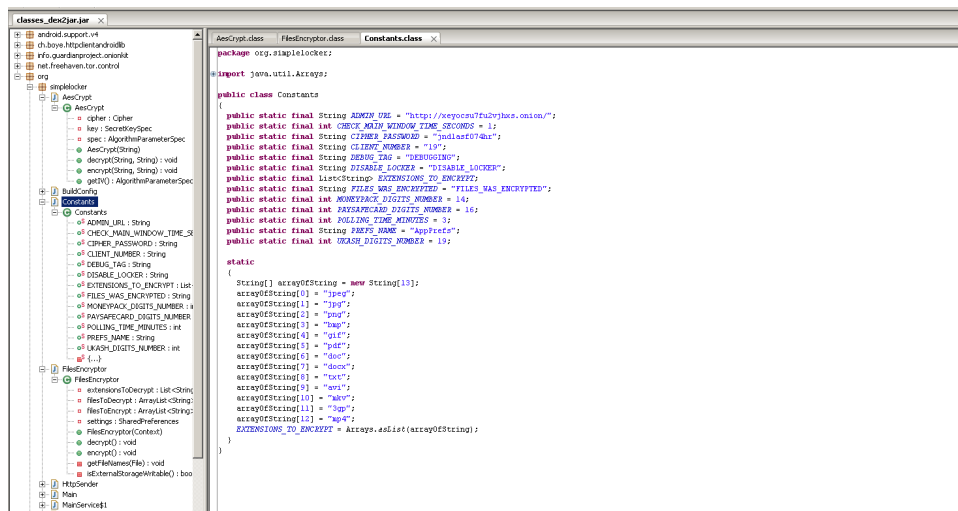


Figure 13: SDK Manager and Test Device

As a first step of the dynamic analysis the Android application is installed with the command 'adb install Android_ransom.apk'. After installing we will make use of "adb shell" command we will enter into the 'root @ generic' and then use 'ps' to obtain the status of the processes after checking the status 'kill PID (process id of simple locker)' and then exit from the "adb shell".

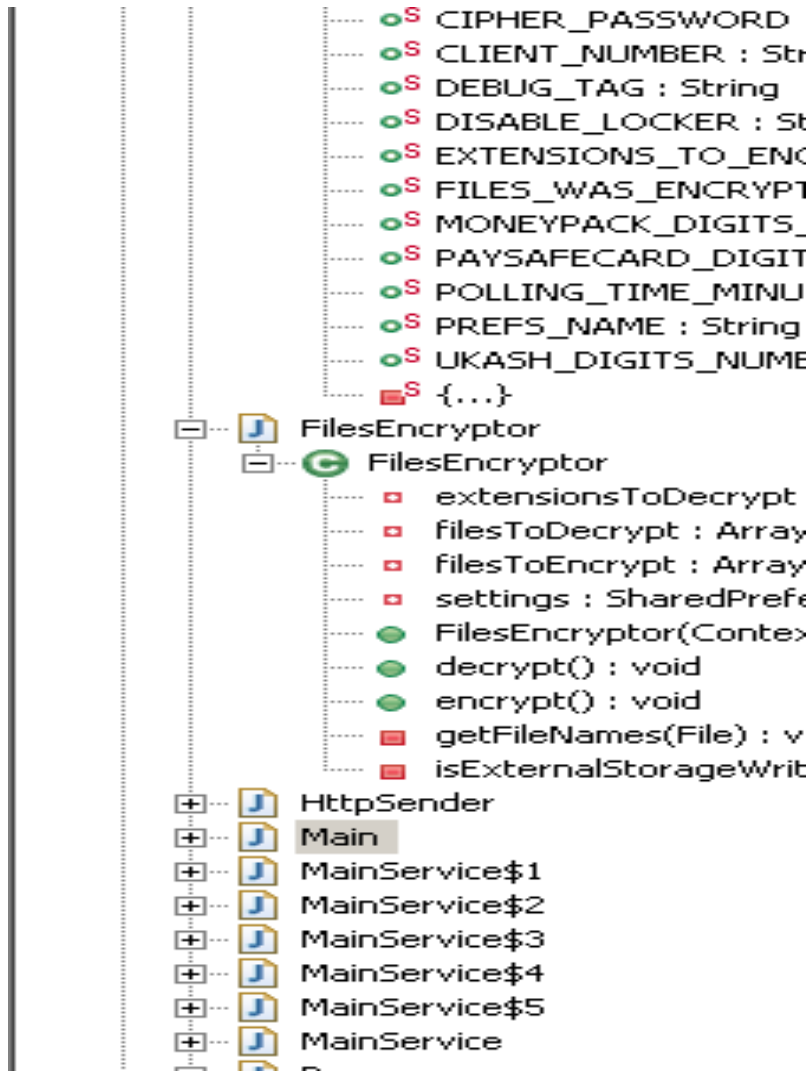


Figure 14: SDK Manager and Test Device

So from the above command we can understand that the processor. SimpleLocker is responsible for encryption of files.

To view the encrypted files again enter the following commands “adb shell” and go to the pictures in the sdcard using “cd /sdcard/pictures” and if we list all the files, we can see that “pic.png.enc” is there in the pictures.

The command “adb uninstall org.simplelocker” is used to uninstall the sexxonix and the result of the this command can be seen in the right side of the below figure

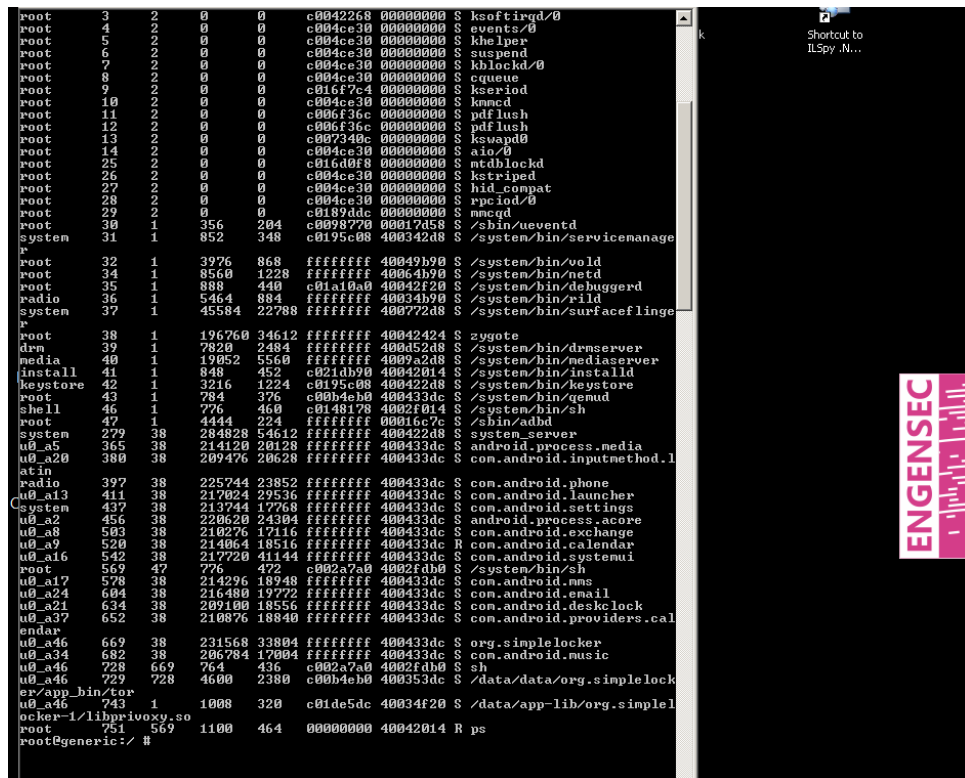


Figure 15: SDK Manager and Test Device

As the previous format of the file is unable to decompile with the java decompiler, so the file is converted to .apk which can be decompiled. To convert the file to .apk the following command is used “apktool.bat d -f Android.ransom.apk Now in the simple locker folder we will go through the



Figure 16: SDK Manager and Test Device

Android.ransom smali and select the Mainservices In that file identify the encrypt() and replace it with decrypt() here Whenever it executes it calls the function it reads the lines

```

C:\WINDOWS\system32\cmd.exe
atform-tools;c:\Android\adt-bundle-windows-x86-20140321\sdk\tools;c:\Python27;c:\
\Android\android-ndk-r9d;c:\cygwin\bin;c:\Android\apktool\1.4.6;c:\Python27\Sc
ip
ts;c:\Python27\;
'Apkset' is not recognized as an internal or external command,
operable program or batch file.

C:\STUDENT LABS\Lab8 - Android Malware Analysis\Ransom>java -jar -Duser.language
=en "c:\Android\apktool\1.4.6\apktool.jar" d -f Android_ransom.apk
I: Using Apktool 2.0.0-Beta9 on Android_ransom.apk
I: Loading resource table...
I: Loading resource table...
I: Decoding AndroidManifest.xml with resources...
I: Loading resource table from file: C:\Documents and Settings\Administrator\apk
tool\framework\1.apk
I: Regular manifest package...
I: Decoding file-resources...
I: Decoding values */* XMLs...
Cleaning up unclosed ZipFile for archive C:\Documents and Settings\Administrator
\apktool\framework\1.apk
I: Baksmaling...
I: Copying assets and libs...
I: Copying unknown files/dir...
I: Copying original files...

C:\STUDENT LABS\Lab8 - Android Malware Analysis\Ransom>

```

Figure 17: SDK Manager and Test Device

Now a new apk file is created by using the following command “apktool.bat b” in the Andriod_random folder which is shown in the below figure.

```

C:\WINDOWS\system32\cmd.exe
atform-tools;c:\Android\adt-bundle-windows-x86-20140321\sdk\tools;c:\Python27;c:\
\Android\android-ndk-r9d;c:\cygwin\bin;c:\Android\apktool\1.4.6;c:\Python27\Sc
ip
ts;c:\Python27\;
'Apkset' is not recognized as an internal or external command,
operable program or batch file.

C:\STUDENT LABS\Lab8 - Android Malware Analysis\Ransom>java -jar -Duser.language
=en "c:\Android\apktool\1.4.6\apktool.jar" d -f Android_ransom.apk
I: Using Apktool 2.0.0-Beta9 on Android_ransom.apk
I: Loading resource table...
I: Loading resource table...
I: Decoding AndroidManifest.xml with resources...
I: Loading resource table from file: C:\Documents and Settings\Administrator\apk
tool\framework\1.apk
I: Regular manifest package...
I: Decoding file-resources...
I: Decoding values */* XMLs...
Cleaning up unclosed ZipFile for archive C:\Documents and Settings\Administrator
\apktool\framework\1.apk
I: Baksmaling...
I: Copying assets and libs...
I: Copying unknown files/dir...
I: Copying original files...

C:\STUDENT LABS\Lab8 - Android Malware Analysis\Ransom>

```

Figure 18: SDK Manager and Test Device

To install the random apk the path need to set to dist folder with using “cd dist”, and then the following command is used to install the application “adb install Android_ransom.apk”. below figure we can see that it is showing the error as the In “INSTALL_PARSE_FAILED_NO_CERTIFICATES” which indicated there is no certificate and failed to install.

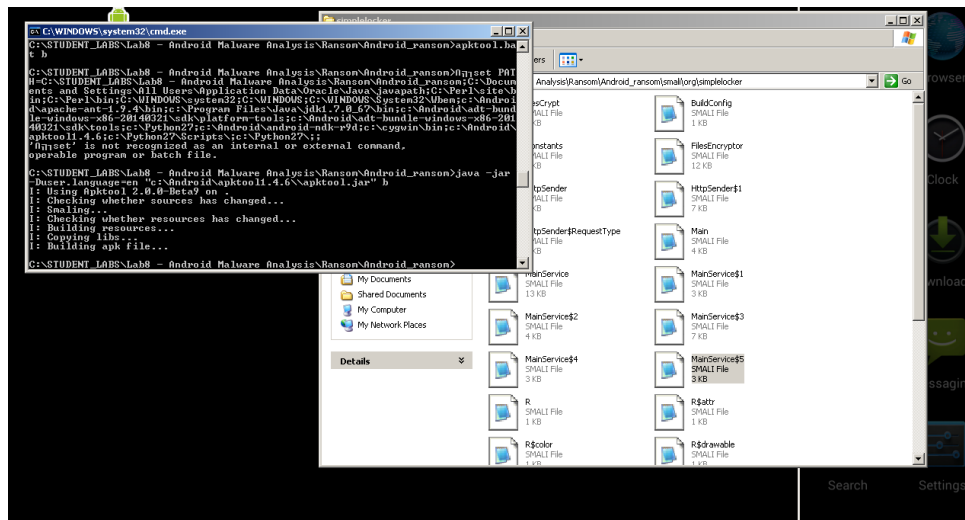


Figure 19: SDK Manager and Test Device

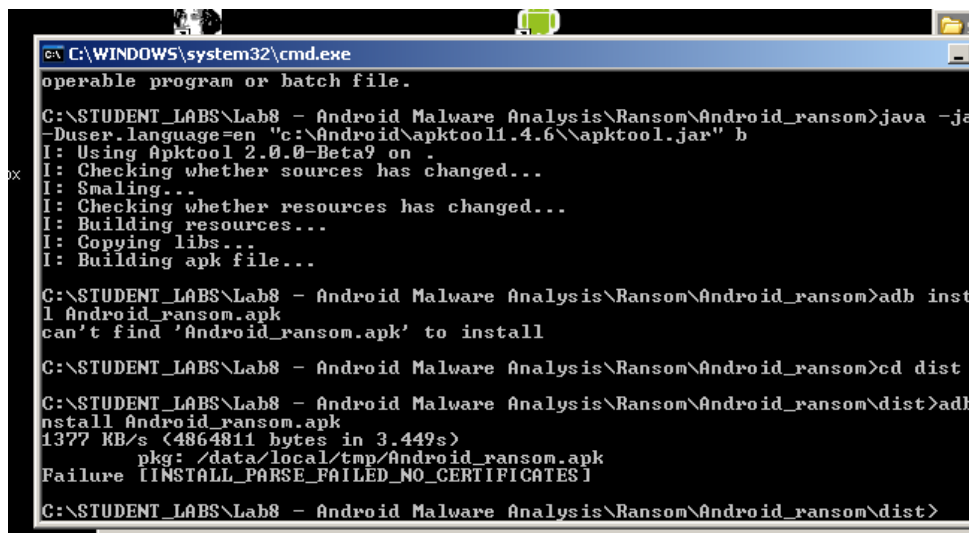


Figure 20: SDK Manager and Test Device

To rectify the no certificate error we will sign a certificate from the “sign +” tool which is found in the Android folder.

The certificate signing is shown in the below figure on the left side and the application used in seen on the right side of the picture.

After signing the certificates we can install the signed apk by using the following command “adb install Android_ransom.signed.apk”

As the above to decrypt the apk we will enter the shell by using “adb shell” and check through the process status with “PS” command and kill the process by using “kill PID” the following command.

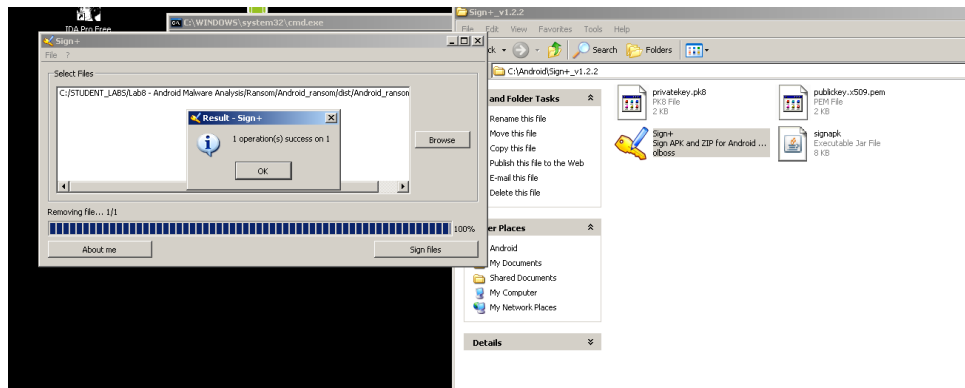


Figure 21: SDK Manager and Test Device

In the below picture 2 observation can be made on the right side we can see that in Android_randomfolder the newly installed signed version file and on the left side we can see the adb shell and the process status.

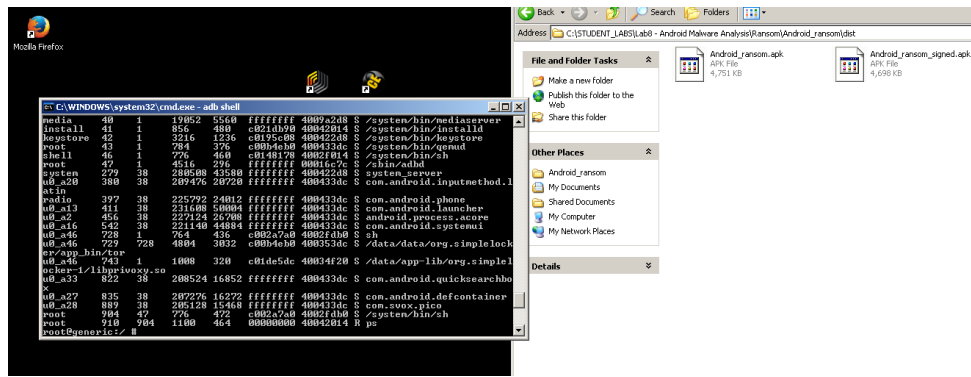


Figure 22: SDK Manager and Test Device

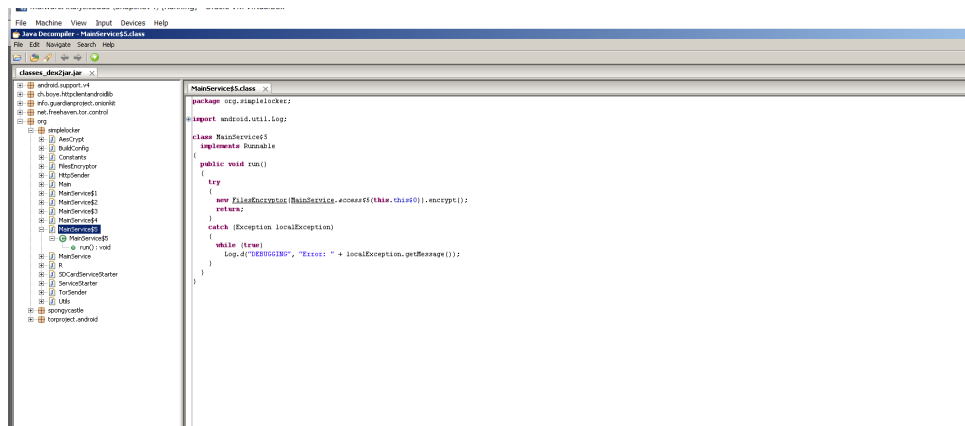


Figure 23: SDK Manager and Test Device

In the below figure we can see we performed the decryption successfully because on the right side of the picture, it is from the gallery. and on the left side when we checked the folder picture in the SD card, we can see that the "pic.png" doesn't have an extension of .enc which prove decryption is successful

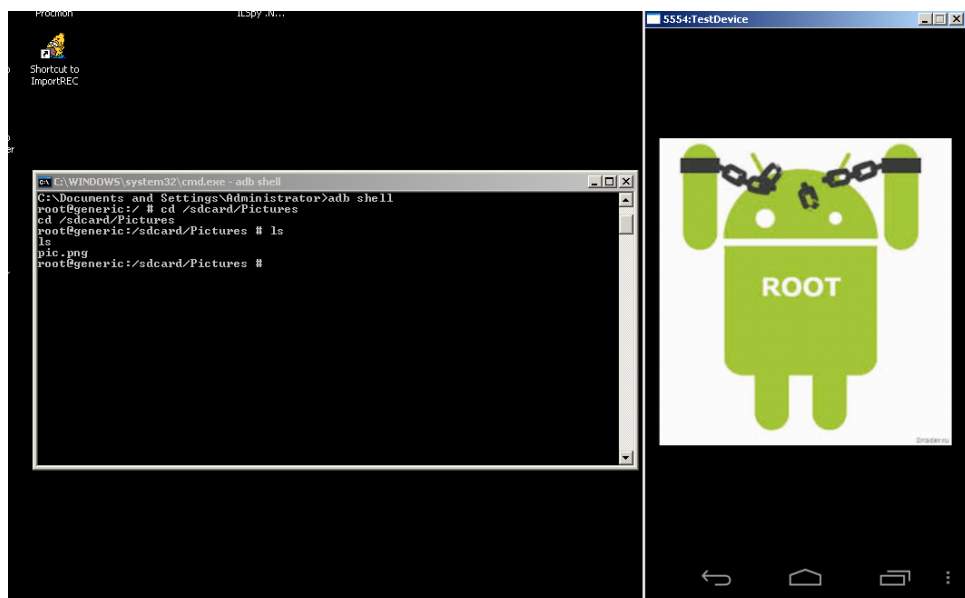


Figure 24: SDK Manager and Test Device